DOMAIN 01

Electronics

Virtual Electronics Project Challenge | 1st Year: idea + circuit | 2nd Year: full simulation in Tinkercad

OBJECTIVE

Using the component list provided, come up with an electronic project. You need not use every component. Use your creativity to build something useful, innovative or interesting. Think beyond the obvious: turn a simple circuit into a meaningful solution.

RULES (BOTH YEARS)

- Any combination of listed components may be used; using all of them is not compulsory.
- The project must have a clear purpose or functionality.
- Everything is done virtually. No physical components, breadboards or jumper wires.
- Resistors and capacitors are not in the list.
- Innovative applications are encouraged over commonly available projects.
- A component outside the list needs the organizers' permission.

1ST YEAR: IDEA AND CIRCUIT ONLY

Bring your **project idea** and the **circuit / schematic** on the interview date. The **report part must be handwritten**. Code and simulation are not required.

2ND YEAR: FULL SIMULATION

Same task, but you must also simulate it in **Tinkercad**. On the interview date, **bring your laptop** with the **circuit**, the **program / code** (if applicable) and the **working simulation**, plus a **handwritten report** with a brief explanation of the working principle. Be ready to open and demonstrate it.

BRING ON INTERVIEW DATE

- **1st year:** handwritten report (idea + circuit/schematic).
- **2nd year:** laptop with the Tinkercad project, and the handwritten report.

COMPONENT LIST

Component	Qty
Arduino UNO	1
LED	10
RGB LED	1
Push Button	4
Buzzer	1
LDR	2
IR Sensor	2
Ultrasonic Sensor (HC-SR04)	1
PIR Sensor	1
DHT11 Temperature & Humidity Sensor	1

Component	Qty
Potentiometer	2
7-Segment Display	1
16x2 LCD Display	1
Servo Motor	1
DC Motor	1
L293D Motor Driver	1
Relay Module	1
BC547 Transistor	3
1N4007 Diode	4